

☒ Easy (10 Questions)

1.

What does DoS stand for?

- A) Denial of Service
- B) Department of Security
- C) Data over Server
- D) Disk Operating System

Answer: A

Explanation: DoS means Denial of Service.

2.

DDoS attacks involve:

- A) A single attacker
- B) Multiple compromised systems
- C) Only email spam
- D) No network involvement

Answer: B

Explanation: DDoS uses many systems.

3.

A botnet is:

- A) A security tool
- B) A network of infected computers controlled by an attacker
- C) A firewall
- D) A type of antivirus

Answer: B

Explanation: Botnets are networks of bots.

4.

Smurf attacks use:

- A) UDP flood
- B) ICMP broadcast requests
- C) TCP reset packets
- D) HTTP GET floods

Answer: B

Explanation: Smurf attacks use ICMP broadcast.

5.

In a SYN flood attack, attacker:

- A) Completes handshake quickly
- B) Sends SYN requests but never completes handshake
- C) Sends only FIN packets
- D) Blocks SYN packets

Answer: B

Explanation: Half-open TCP connections exhaust resources.

6.

Session hijacking means:

- A) Blocking a user session
- B) Taking control of a legitimate session
- C) Logging off users
- D) Encrypting sessions

Answer: B

Explanation: Hijacking means takeover.

7.

Spoofing involves:

- A) Stealing session cookies
- B) Faking IP or MAC addresses
- C) Encrypting data
- D) Blocking traffic

Answer: B

Explanation: Spoofing falsifies identities.

8.

Which attack floods victim with traffic from many machines?

- A) DoS
- B) DDoS
- C) Phishing
- D) Keylogging

Answer: B

Explanation: DDoS uses many machines.

9.

Which of these is a step in session hijacking?

- A) Password cracking
- B) Session identification
- C) Virus scanning
- D) Firewall configuration

Answer: B

Explanation: Session hijacking starts by finding session IDs.

10.

A countermeasure for SYN flood is:

- A) Use SYN cookies
- B) Disable firewalls
- C) Use Telnet
- D) Open all ports

Answer: A

Explanation: SYN cookies prevent resource exhaustion.

☒ Medium (20 Questions)

11.

How do botnets communicate with their controller?

- A) Email only
- B) Command and Control (C&C) servers
- C) Social media
- D) USB drives

Answer: B

Explanation: Botnets use C&C servers.

12.

What is a typical effect of a Smurf attack?

- A) Website defacement
- B) Overloading victim with ICMP replies
- C) Password theft
- D) Virus infection

Answer: B

Explanation: Amplified ICMP traffic overwhelms victim.

13.

Which layer does SYN flood attack target?

- A) Application layer
- B) Transport layer (TCP)
- C) Network layer
- D) Physical layer

Answer: B

Explanation: SYN floods exploit TCP handshake.

14.

IP spoofing is used in:

- A) Firewall configuration
- B) Hiding the attacker's true IP address
- C) Encrypting packets
- D) User authentication

Answer: B

Explanation: Spoofing masks the attacker.

15.

Which flag in TCP handshake is sent first?

- A) SYN
- B) ACK
- C) FIN
- D) RST

Answer: A

Explanation: SYN initiates the connection.

16.

Which of the following is NOT a session hijacking type?

- A) Passive hijacking
- B) Active hijacking
- C) Denial hijacking
- D) MITM attack

Answer: C

Explanation: Denial hijacking is not a term.

17.

Session fixation attack involves:

- A) Guessing passwords
- B) Forcing user to use attacker's session ID
- C) Sniffing passwords
- D) Brute force attacks

Answer: B

Explanation: Attacker fixes a session ID.

18.

Which cookie attribute helps protect from XSS?

- A) HttpOnly
- B) Secure
- C) Path
- D) Domain

Answer: A

Explanation: HttpOnly prevents client-side script access.

19.

Which attack uses many infected machines to overwhelm the target?

- A) Virus attack
- B) DDoS attack
- C) Brute force
- D) Trojan horse

Answer: B

Explanation: DDoS uses multiple machines.

20.

Which tool can detect session hijacking?

- A) IDS/IPS
- B) FTP
- C) SSH
- D) DNS server

Answer: A

Explanation: Intrusion detection systems spot anomalies.

21.

A botnet controller is also called:

- A) Rootkit

- B) Command and Control (C&C) server
- C) Firewall
- D) Proxy server

Answer: B

Explanation: C&C server controls bots.

22.

Which is a key symptom of a SYN flood attack?

- A) Large number of half-open TCP connections
- B) Sudden drop in DNS queries
- C) Password leaks
- D) Web defacement

Answer: A

Explanation: SYN flood creates half-open connections.

23.

What is the goal of session hijacking?

- A) Shutdown system
- B) Gain unauthorized access to session data
- C) Encrypt communications
- D) Block connections

Answer: B

Explanation: Hijacking aims to control sessions.

24.

Which step involves guessing session tokens?

- A) Session sniffing
- B) Session prediction
- C) Session fixation
- D) Session injection

Answer: B

Explanation: Prediction guesses tokens.

25.

What technique prevents session hijacking by verifying client IP and user-agent?

- A) Encryption
- B) Session binding
- C) Spoofing
- D) Port scanning

Answer: B

Explanation: Session binding ties session to client.

26.

Which of these is NOT a characteristic of a botnet?

- A) Distributed
- B) Controlled by C&C
- C) Always visible on network
- D) Used for DDoS attacks

Answer: C

Explanation: Botnets try to remain hidden.

27.

Which method can reduce impact of a DDoS?

- A) Traffic filtering and rate limiting
- B) Disabling firewalls
- C) Using default passwords
- D) Opening all ports

Answer: A

Explanation: Filtering reduces attack traffic.

28.

Which TCP flag combination indicates a half-open connection?

- A) SYN sent, no ACK received
- B) SYN and ACK both received
- C) FIN sent
- D) RST sent

Answer: A

Explanation: No ACK means half-open.

29.

Which is a common technique used in DDoS amplification?

- A) IP Spoofing
- B) Phishing
- C) Password cracking
- D) SQL Injection

Answer: A

Explanation: Spoofed IP amplifies traffic.

30.

How can HTTPS prevent session hijacking?

- A) Encrypts session tokens and data
- B) Disables cookies
- C) Changes IP address
- D) Blocks SYN packets

Answer: A

Explanation: Encryption protects tokens.

☒ Hard (10 Questions)

31.

In a Smurf attack, what network device setting helps prevent amplification?

- A) Disable IP-directed broadcasts

- B) Enable port forwarding
- C) Disable DHCP
- D) Enable NAT

Answer: A

Explanation: Blocking directed broadcasts prevents smurf amplification.

32.

Which of the following is a known countermeasure for SYN flood attacks?

- A) Increasing the SYN backlog queue size
- B) Disabling TCP
- C) Using UDP instead of TCP
- D) Using Telnet

Answer: A

Explanation: Larger backlog helps handle half-open connections.

33.

Which type of hijacking involves altering the data between two parties?

- A) MITM attack
- B) Passive hijacking
- C) Session fixation
- D) Password spraying

Answer: A

Explanation: MITM intercepts and alters data.

34.

What is the primary purpose of a Trojan horse in a botnet?

- A) To spread spam
- B) To infect and control machines as bots
- C) To encrypt files
- D) To deface websites

Answer: B

Explanation: Trojans recruit bots.

35.

Which of the following best describes 'session injection'?

- A) Guessing session tokens
- B) Injecting malicious packets into a session
- C) Logging out users
- D) Phishing for credentials

Answer: B

Explanation: Session injection sends malicious packets.

36.

What is a common indication of session fixation?

- A) User forced to use a pre-set session ID
- B) User changes password frequently
- C) Use of HTTPS only
- D) No cookies sent

Answer: A

Explanation: Attacker sets session ID.

37.

Which protocol flag is often abused in stealth scans?

- A) FIN
- B) SYN
- C) ACK
- D) PSH

Answer: A

Explanation: FIN scan tries to bypass firewall.

38.

How does a botnet avoid detection?

- A) Using encrypted C&C communication
- B) Only attacking once a year
- C) Broadcasting openly
- D) Using static IPs

Answer: A

Explanation: Encryption hides control messages.

39.

What is a major limitation of session hijacking?

- A) Requires session ID or token knowledge
- B) Can be done without network access
- C) Doesn't need encryption
- D) Does not rely on cookies

Answer: A

Explanation: Hijacker needs session info.

40.

Which of the following techniques best mitigates session hijacking?

- A) Using secure cookies with HttpOnly and Secure flags
- B) Using HTTP only
- C) Disabling firewalls
- D) Using Telnet for sessions

Answer: A

Explanation: Secure and HttpOnly flags protect cookies.